ටිංටිං (TinTin)
Jump. Dodge. Survive.

SE4031 – Games Development
Assignment 02 (Individual)

Student Name	Fernando W W R D
IT Number	IT22361554
Module	SE4031 – Games Development
Assignment	Assignment 02 (Individual)
Game Title	ටිංටිං TinTin
Game Engine	Unity 6000.2.6f1
Language	C#
Platform	Windows PC (.exe build)
Submission Year	2026

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1. Game Title

TinTin

Why "TinTin"?

"TinTin" is a name Sri Lankan children commonly give to squirrels, a term of endearment used across the island when spotting the little creatures darting through garden trees and along compound walls. The name was chosen not from a game design brief but from a real moment: the developer spotted a squirrel on a mango tree outside their home in Moratuwa, and the idea for the game was born in that instant.

The game is intentionally Sri Lankan in spirit:
The setting is a tropical neighborhood garden wall; brick walls, mango trees, palm-lined backgrounds.
The obstacles are broken glass shards embedded in walls; a common sight on boundary walls across Sri Lanka, used as a security measure.
The collectible is a mango ; not a coin, not a star, but a mango. Because that is what TinTin was after.
The enemies are crows ; the most common garden bird in Sri Lanka and the natural rival of any fruit-stealing squirrel.

Together, TinTin is a game that could only have come from this place. It is small, personal, and unmistakably Sri Lankan.

2. Student Details

Field	Details
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3. Summary of the Game Concept

TinTin is a 2D endless side-scrolling runner game built in Unity 6, set in a tropical Sri Lankan neighborhood. The player controls TinTin, a small squirrel, as he races endlessly along the top of a brick garden wall, jumping over obstacles, collecting mangoes to restore health, and throwing mango seeds at enemy crows that try to stop him.

The game draws its core mechanics from the endless runner genre but wraps them in an original Sri Lankan cultural context. The obstacles are tree branches hanging down from above, requiring TinTin to jump through gaps. The wall below is lined with broken glass shards, a familiar Sri Lankan security feature, which serve as the bottom boundary. Mangoes hang from trees as health pickups, and crows patrol the wall as enemies.

Core Gameplay Loop

- TinTin runs automatically ,the world scrolls left continuously.
- Press Space to jump over obstacles ,passing through the gap scores +1 point.
- Hit an obstacle, health decreases, TinTin passes through.
- Collect mangoes on the wall health restores up to maximum.
- Enemy crows patrol the wall ,run into one and health decreases.
- Left click to throw a mango seed ,destroys a crow and scores +1 point.
- Speed increases over time ,the game gets progressively faster and harder.
- Health reaches zero ,Game Over screen appears with final score and high score.

Unique Features

- Sri Lankan cultural theme ,brick walls, broken glass obstacles, mangoes, crows, tropical garden background.
- Animated squirrel character ,6-frame animation set including run and jump cycles with sprites generated with AI and animated in Unity.
- Dynamic health bar gradient ,color updates every frame from green to yellow to red based on exact health value.
- Camera shake on enemy collision ,physical impact feedback using screen shake coroutine.
- High score persistence ,best score saved between sessions using PlayerPrefs.

4. Description of Core Features

4.1 Health System (Requirement 1)

TinTin starts with 100 health points displayed as a UI Slider bar in the top left of the screen. The health bar fill color smoothly interpolates from green (full health) through yellow (mid health) to red (critical) every frame using Color.Lerp() in HealthManager.UpdateHealthUI(). This color update is continuous and frame-accurate ,not threshold-based.

Each collision with a branch obstacle decreases health by 5 points. Each collision with an enemy crow decreases health by 10 points. TinTin physically passes through obstacles on collision rather than stopping. When health reaches zero, GameManager.GameOver() is called: Time.timeScale is set to 0 and the Game Over panel is displayed. Health resets to maxHealth (100) on restart via SceneManager.LoadScene().

4.2 Score System (Requirement 2)

Score is displayed as a TextMeshPro UI element at the top of the screen. A ScoreTrigger child object with a Box Collider 2D (Is Trigger enabled) sits inside the gap of each obstacle. When TinTin passes cleanly through the gap, OnTriggerEnter2D fires in ScoreTrigger.cs and calls ScoreManager.Instance.AddScore(1), but only if TinTin's vertical position is confirmed to be within the gap boundaries ,preventing score from registering if TinTin clipped an obstacle edge. Destroying a crow enemy with a mango seed projectile awards +1 score. Score resets to 0 on restart. High score is saved and displayed on the Game Over screen using PlayerPrefs.

4.3 Health Packs (Requirement 3)

Mangoes serve as health packs. They spawn on top of the scrolling brick wall at regular intervals via MangoSpawner.cs, positioned at ground level so TinTin collects them naturally while running ,no jump required. Each mango has a Circle Collider 2D (Is Trigger), the Mango tag, and MangoPickup.cs. On trigger collision with TinTin, HealHealth(20f) is called (capped at maxHealth = 100) and the mango is destroyed immediately. If TinTin misses a mango, Destroy(gameObject, autoDestroyTime) in Start() auto-destroys it after a fixed time interval. Mangoes are visually distinct from crows ,bright orange/yellow rounded fruit vs dark angular bird.

4.4 Projectile Attacks (Requirement 4)

Left mouse click instantiates the mango seed Projectile prefab at a position slightly in front of TinTin's current world position (offset +1.5f on X axis). The projectile has a Rigidbody2D (Body Type: Kinematic, gravity 0), Circle Collider 2D (Is Trigger), and moves rightward at a fixed speed in Projectile.Update(). It auto-destroys after 2 seconds via Destroy(gameObject, destroyAfterTime). On trigger collision with an Enemy-tagged object, ScoreManager.Instance.AddScore(1) is called, the enemy is destroyed, and the projectile is destroyed. On collision with anything else the projectile is destroyed immediately.

4.5 Enemy Mechanics (Requirement 5)

Enemy crows spawn on top of the scrolling brick wall at regular intervals via EnemySpawner.cs, positioned slightly above the wall surface. Each crow has a Circle Collider 2D (Is Trigger), the Enemy tag, and EnemyBird.cs. On collision with a mango seed projectile: ScoreManager.Instance.AddScore(1) is called, both the projectile and the crow are destroyed. On collision with TinTin: HealthManager.Instance.TakeDamage(10f) is called, CameraShake.Instance.Shake() is triggered, and the crow is destroyed. Crows auto-destroy after a fixed time via Destroy(gameObject, autoDestroyTime) if they scroll off the left side of the screen. Crows are visually distinguishable from mangoes by both color (dark grey/black vs bright orange) and shape (bird silhouette vs round fruit).

4.6 Speed Increase Over Time (Requirement 6)

Game speed is managed centrally by GameSpeedManager.cs using a static Instance pattern. Starting speed is set at 3f and increases by speedIncreaseRate * Time.deltaTime every frame, capped at maxSpeed of 8f. All scrolling objects ,background, ground, obstacles, mangoes, and enemies ,read their movement speed from GameSpeedManager.Instance.CurrentSpeed multiplied by their individual speedMultiplier. The background scrolls at 0.4x the game speed to create a parallax depth effect. The ground and all gameplay objects scroll at 1x game speed. Speed is reset to startSpeed on restart via GameSpeedManager.ResetSpeed().

4.7 Creative Feature (Requirement 7)

The creative feature implemented is a dynamic gradient health bar that updates continuously every frame based on TinTin's exact current health ratio.

Implementation in HealthManager.UpdateHealthUI():

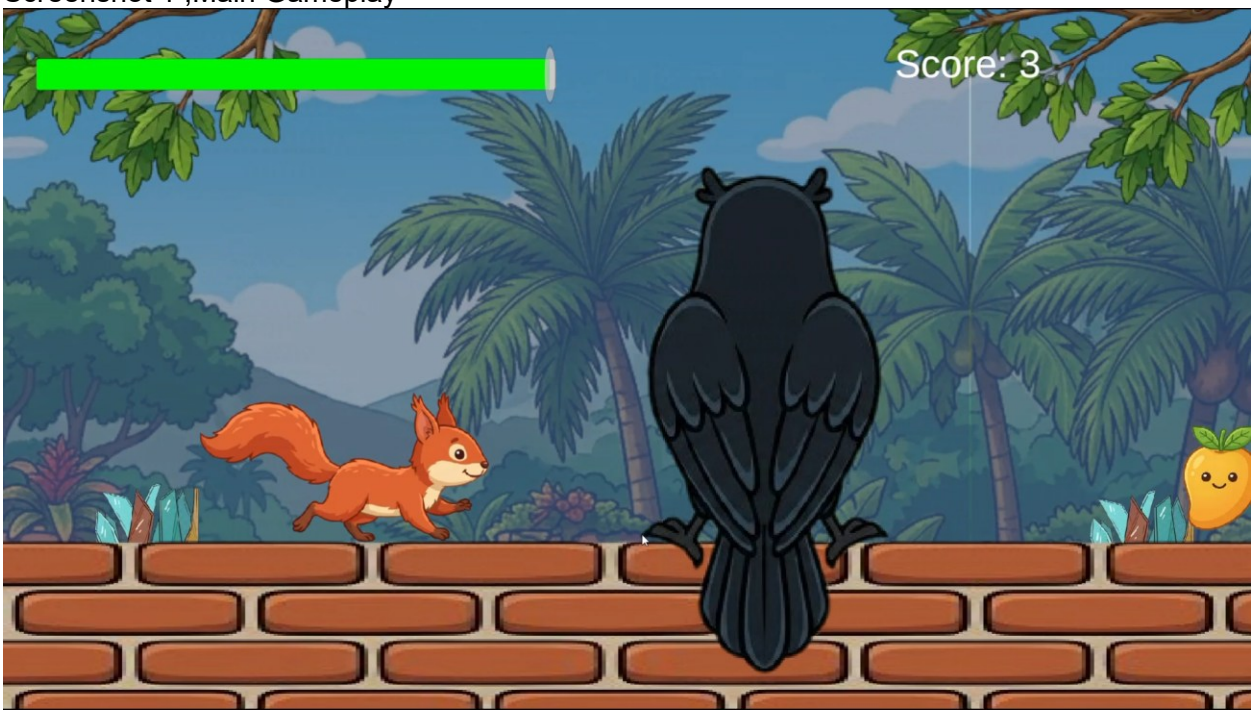
```
float healthPercent = currentHealth / maxHealth;
healthBarFill.color = Color.Lerp(Color.red, Color.green, healthPercent);
```

The color is recalculated and applied to the health bar fill Image component on every single frame. At 100% health the bar is bright green. At 50% health it is yellow. At 0% health it is bright red. The transition is perfectly smooth and continuous ,never jumping between fixed threshold colors. This provides instant visual health awareness and reinforces the danger of each hit in a way a static color could not.

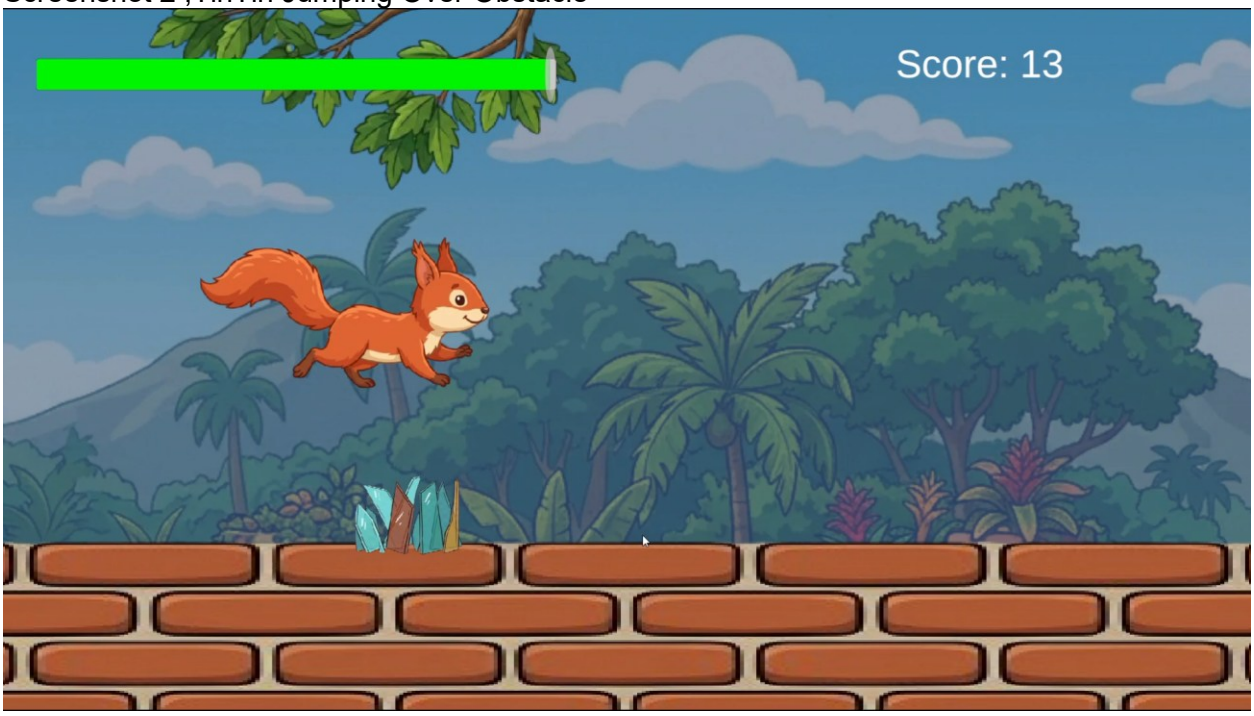
Additionally, camera shake was implemented as a bonus visual enhancement. When TinTin collides with an enemy crow, CameraShake.Instance.Shake() is called. The Main Camera's localPosition is randomly offset using Random.insideUnitCircle * shakeMagnitude every frame for shakeDuration seconds, then returned to its exact original position. This creates immediate physical impact feedback for every enemy collision.

5. Screenshots of Gameplay

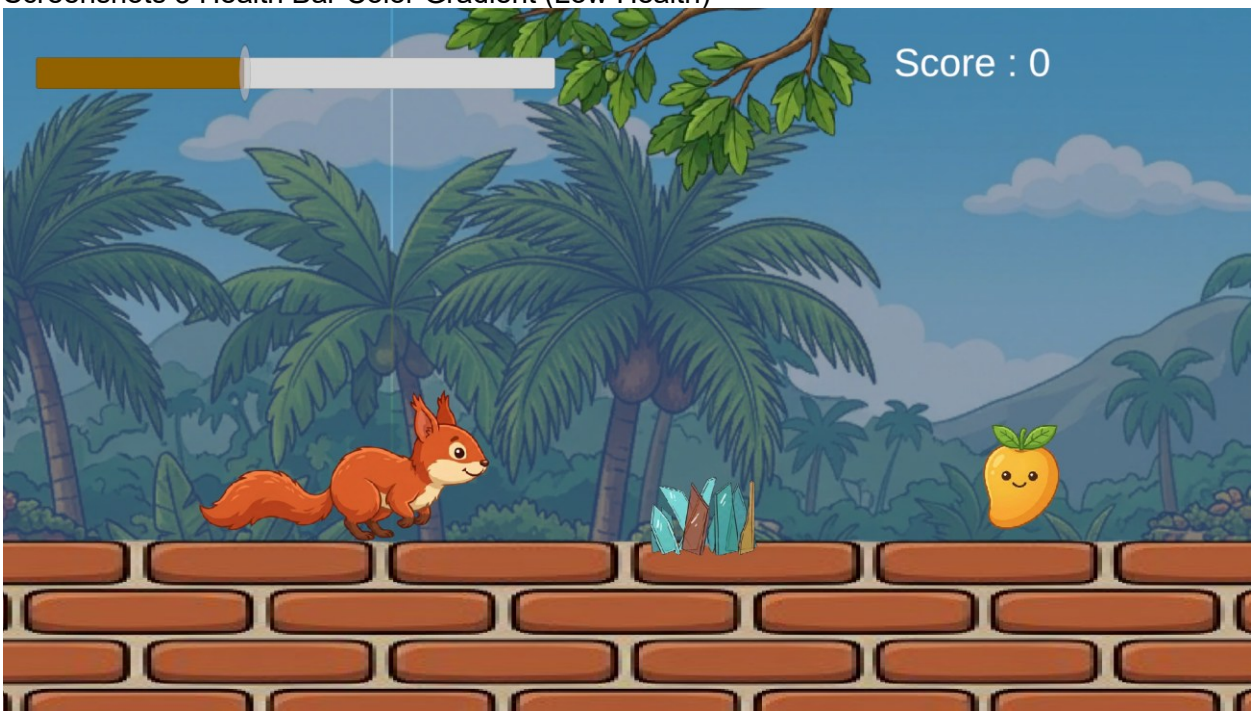
Screenshot 1 ,Main Gameplay

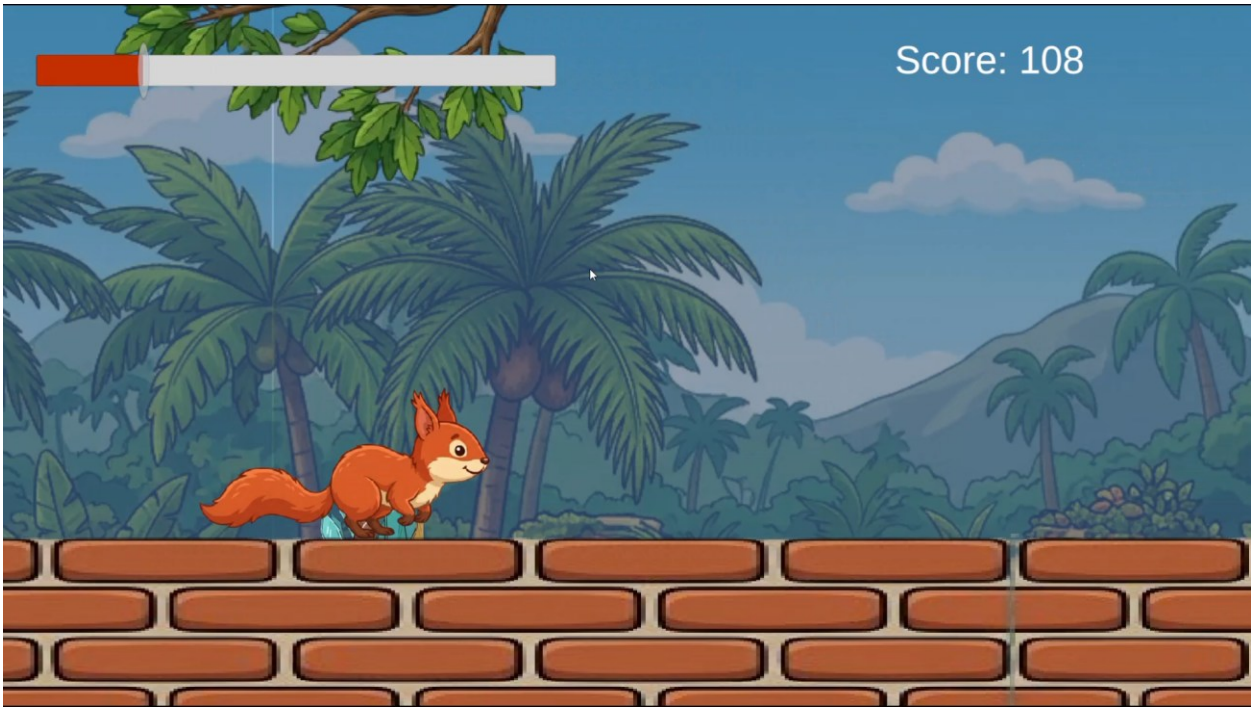


Screenshot 2 ,TinTin Jumping Over Obstacle

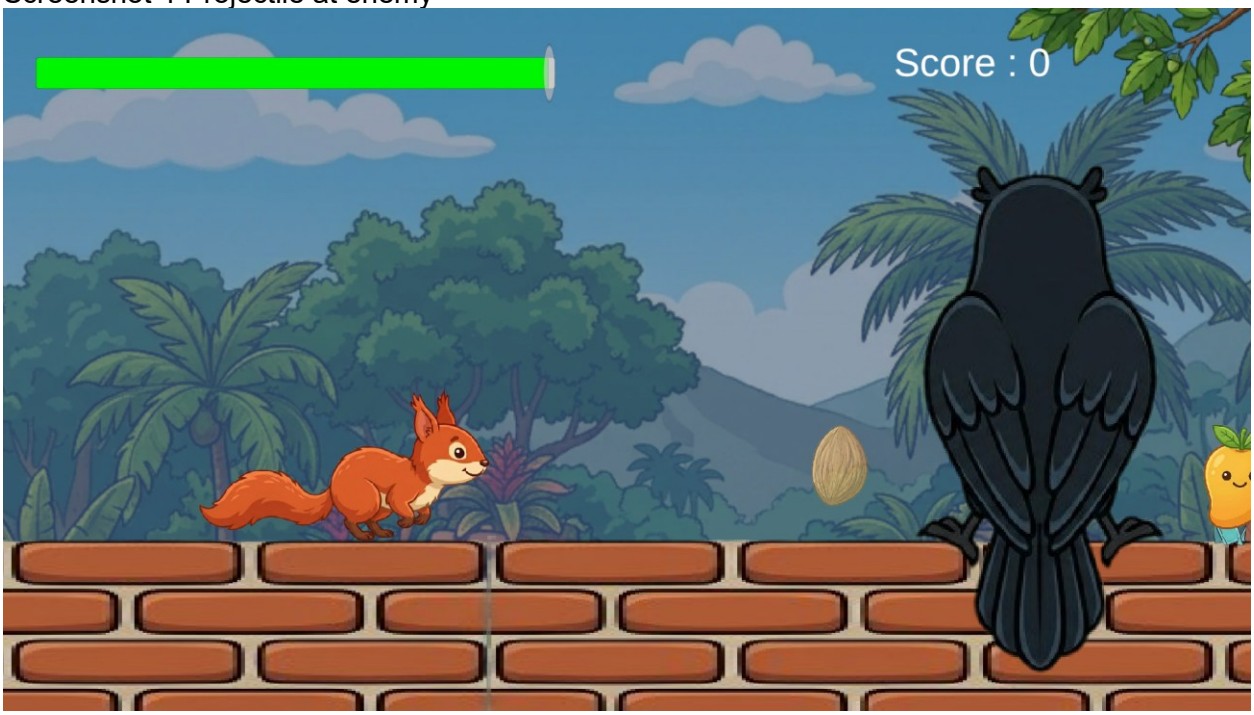


Screenshots 3 Health Bar Color Gradient (Low Health)





Screenshot 4 Projectile at enemy



Screenshot 4 Game Over Screen

6. Control Guide

Input	Action
Space Bar	Jump ,TinTin jumps upward to clear obstacles
Left Mouse Button	Throw mango seed projectile from TinTin's position
Restart Button (UI)	Restart game after Game Over ,resets health, score, and speed

TinTin uses an intentionally simple two-action control scheme. TinTin runs automatically ,the player only needs to jump and shoot. The challenge comes entirely from timing, spatial awareness, and resource management rather than complex inputs, keeping the game accessible while remaining demanding at higher speeds.

Gameplay Tips

- Time your jump carefully ,the gap between branch and ground requires a well-timed press, not an early one.
- Collect mangoes whenever possible ,health is your only resource and restoring even a small amount matters.
- Shoot crows before they reach TinTin ,+2 score and zero damage is always the better trade.
- Watch the health bar color ,when it turns yellow, prioritize mango collection over scoring.
- Speed increases gradually ,the first 20 seconds are the safest window to build up score.

7. Creative Feature Description

Dynamic Gradient Health Bar

The health bar fill color is recalculated and updated on every single frame based on TinTin's exact current health ratio. Unlike a threshold approach (e.g. turn red below 33%), the gradient is continuous and always perfectly reflects the exact health state.

Code implementation in HealthManager.cs:

```
void UpdateHealthUI()
{
    if (healthBar != null)
        healthBar.value = currentHealth / maxHealth;

    if (healthBarFill != null)
    {
        float healthPercent = currentHealth / maxHealth;
        healthBarFill.color = Color.Lerp(Color.red, Color.green, healthPercent);
    }
}
```

Result: At 100/100 health the bar is bright green. At 50/100 it is yellow. At 10/100 it is deep red. The transition is perfectly smooth and continuous, updating every frame ,not at fixed thresholds.

Camera Shake on Enemy Collision (Bonus Enhancement)

Every time TinTin collides with an enemy crow, the Main Camera runs a screen shake coroutine via CameraShake.cs. The camera's localPosition is randomly offset using Random.insideUnitCircle every frame for 0.2 seconds, then returned to its exact original localPosition. The effect is triggered by EnemyBird.OnTriggerEnter2D() calling CameraShake.Instance.Shake().

Code implementation in CameraShake.cs:

```
IEnumerator DoShake()
{
    float elapsed = 0f;
    while (elapsed < shakeDuration)
    {
        transform.localPosition = originalPos +
            (Vector3)Random.insideUnitCircle * shakeMagnitude;
        elapsed += Time.deltaTime;
        yield return null;
    }
    transform.localPosition = originalPos;
}
```

This creates immediate visceral physical feedback for every enemy hit, making each collision feel impactful rather than just a silent health decrement.

8. Assets Used

8.1 Player Character ,TinTin the Squirrel

Property	Details
Source	AI Generated ,Google Gemini Image Generation
Sprites Created	6 frames: Running 1, Running 2, Prep/Landing, Jumping Up, Fully Stretched, Landed
Background Removal	remove.bg (automated background removal tool)
Animation	Animated in Unity Animator ,Run cycle (frames 1-2 loop) and Jump cycle (frames 3-6)
Collider	Capsule Collider 2D fitted to squirrel body
Script	TinTinController.cs ,handles jump, shoot, collision, animation states
Tag	Player

8.2 Enemy ,Crow

Property	Details
Source	AI Generated ,Google Gemini Image Generation
Background Removal	remove.bg
Visual Style	Dark crow bird sprite ,clearly distinct from mango health pack by color and shape
Collider	Circle Collider 2D (Is Trigger)
Script	EnemyBird.cs ,handles collision with projectile and player, auto-destroy
Tag	Enemy
Behaviour	Scrolls left on wall, deals 10 damage on player contact, destroyed by mango seed (+1 score)

8.3 Health Pack ,Mango

Property	Details
Source	AI Generated ,Google Gemini Image Generation
Background Removal	remove.bg
Visual Style	Bright orange/yellow cartoon mango ,clearly distinct from crow enemy
Collider	Circle Collider 2D (Is Trigger)
Script	MangoPickup.cs ,handles healing, auto-destroy after fixed time
Tag	Mango
Behaviour	Spawns on wall surface, restores 20 health on collection (capped at 100), auto-destroys after 5 seconds

8.4 Projectile ,Mango Seed

Property	Details
Source	AI Generated ,Google Gemini Image Generation
Background Removal	remove.bg
Visual Style	Small mango seed/pip sprite ,distinct from full mango collectible
Collider	Circle Collider 2D (Is Trigger)
Physics	Rigidbody2D, Body Type: Kinematic, gravity 0, moves right at fixed speed
Script	Projectile.cs ,handles rightward movement, collision, auto-destroy after 2 seconds
Tag	Projectile
Behaviour	Fires right on left-click, destroys crow enemy (+1 score), auto-destroys after 2 seconds

8.5 Obstacles ,Tree Branches

Property	Details
Source	AI Generated ,Google Gemini Image Generation (2 variations)
Background Removal	remove.bg
Visual Style	Green leafy branches hanging from above ,2 sprite variations for variety
Collider	Box Collider 2D (Is Trigger)
Script	ScrollingObject.cs ,reads speed from GameSpeedManager, self-destroys off screen
Tag	Obstacle
Behaviour	Spawns from right at random heights, scrolls left, gap at bottom for TinTin to jump through

8.6 Ground ,Brick Wall with Glass Shards

Property	Details
Source	AI Generated ,Google Gemini Image Generation
Background Removal	remove.bg
Visual Style	Sri Lankan-style brick garden wall with colorful broken glass shards on top ,a common security feature on boundary walls across Sri Lanka
Collider	Box Collider 2D for ground physics
Script	BackgroundLooper.cs ,tiles and scrolls continuously at game speed
Sorting Layer	Ground

8.7 Background

Property	Details
Source	AI Generated ,Google Gemini Image Generation
Visual Style	Tropical Sri Lankan garden ,palm trees, mountains, lush greenery, warm sky
Script	BackgroundLooper.cs ,scrolls at 0.4x game speed for parallax depth effect
Sorting Layer	Background (Order -1)

8.8 Game Over Screen

Property	Details
Source	AI Generated ,Google Gemini Image Generation
Visual Style	Mango tree frame, TinTin and crow illustrated, GAME OVER text, Score and Highest Score placeholders, TRY AGAIN button area
Implementation	Used as UI Panel background image in Canvas, score and high score TextMeshPro elements overlaid on designated areas

9. Credits

Game Engine and Development Tools

Unity 6000.2.6f1 ,Game development engine
C# ,Primary programming language
Visual Studio Code ,Code editor
Unity TextMeshPro ,UI text rendering
GitHub Desktop ,Version control and assignment submission

AI Tools Used for Asset Generation

Google Gemini Image Generation ,All game sprites generated including TinTin character frames, mango, mango seed, crow, tree branches, brick wall, glass shards, background, and Game Over screen
remove.bg ,Automated background removal for all generated sprites to produce transparent PNG files

Animation

All animations created by the developer in Unity Animator using AI-generated sprite frames ,no external animation assets used
Run cycle: frames 1 and 2 looped at 12 samples per second
Jump cycle: frames 3 through 6 triggered on Space press, returns to run on landing

References

Unity Manual ,https://docs.unity3d.com/Manual/Quickstart2DCreate.html
Unity Scripting API ,https://docs.unity3d.com/ScriptReference/
Unity TextMeshPro Documentation
,https://docs.unity3d.com/Packages/com.unity.textmeshpro@3.0/manual/
Unity Animator Documentation ,https://docs.unity3d.com/Manual/class-AnimatorController.html

10. Gameplay Demo Video

Item	Link / Details
Gameplay Demo Video	Demo Video Google Drive Link
Unity Project (Google Drive)	Unity Project Zipped Folder Drive Link

The gameplay demo video demonstrates all required features including:

- TinTin running and jumping through obstacle gaps ,score incrementing
- Health bar decreasing on obstacle and enemy collision ,color gradient from green to red
- Mango health packs restoring health
- Mango seed projectile fired on left click ,destroying enemy crow and incrementing score
- Camera shake on enemy collision
- Speed increase over time ,noticeable acceleration
- Game Over screen appearing on health reaching zero
- Final score and high score displayed
- Restart button resetting all systems correctly